UNIDADE LOCAL DE SAÚDE
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Minor Trauma Emergency Circuit (MonoTrauma) Redesign Approach

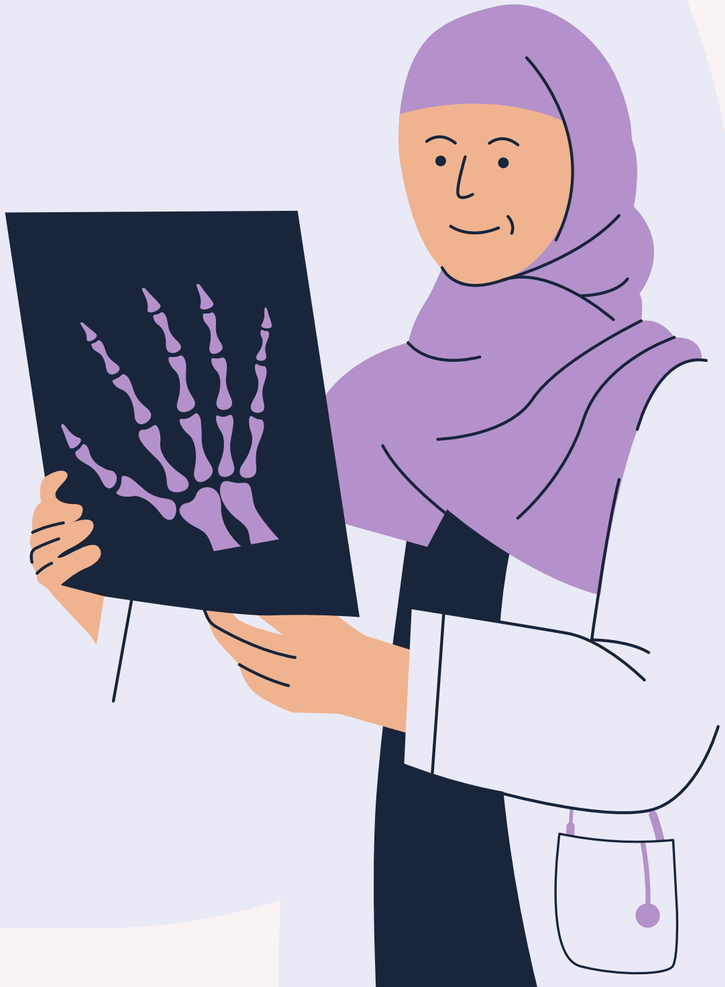
Context & Objectives

The Problem: Increasing pressure on emergency services; need to optimize patient flow and resource utilization.

The Process: Minor Trauma Emergency Circuit "MonoTrauma" – A distinct pathway for minor orthopedic trauma (limbs/vertebral) designed to reduce waiting times

Project Objectives:

-Model the AS-IS process.
-Identify inefficiencies (Qualitative & Quantitative Analysis).
-Propose an improved TO-BE model supported by simulation and financial evaluation



Theoretical Background (Methodology)

Framework: Full BPM Lifecycle (Identification→Redesign)

Modelling: BPMN 2.0 standards using Bizagi Modeler

Qualitative Analysis:

- Value-Added Analysis (VA, BVA, NVA)
- Waste Analysis (Move, Hold, Over-do)
- Issue Register (Impact assessment)
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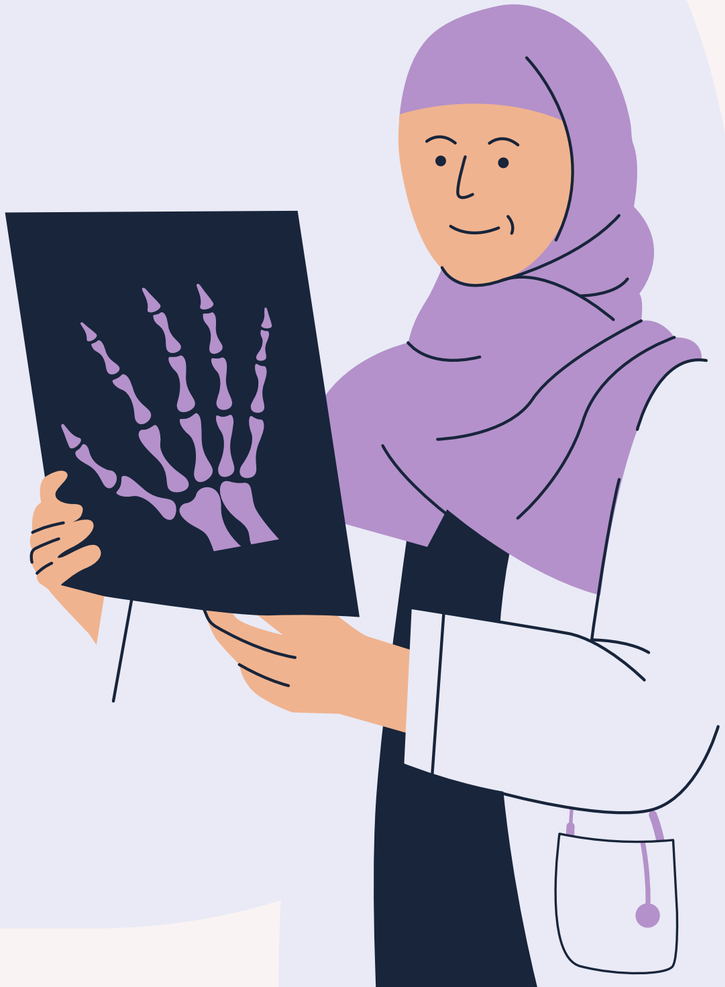
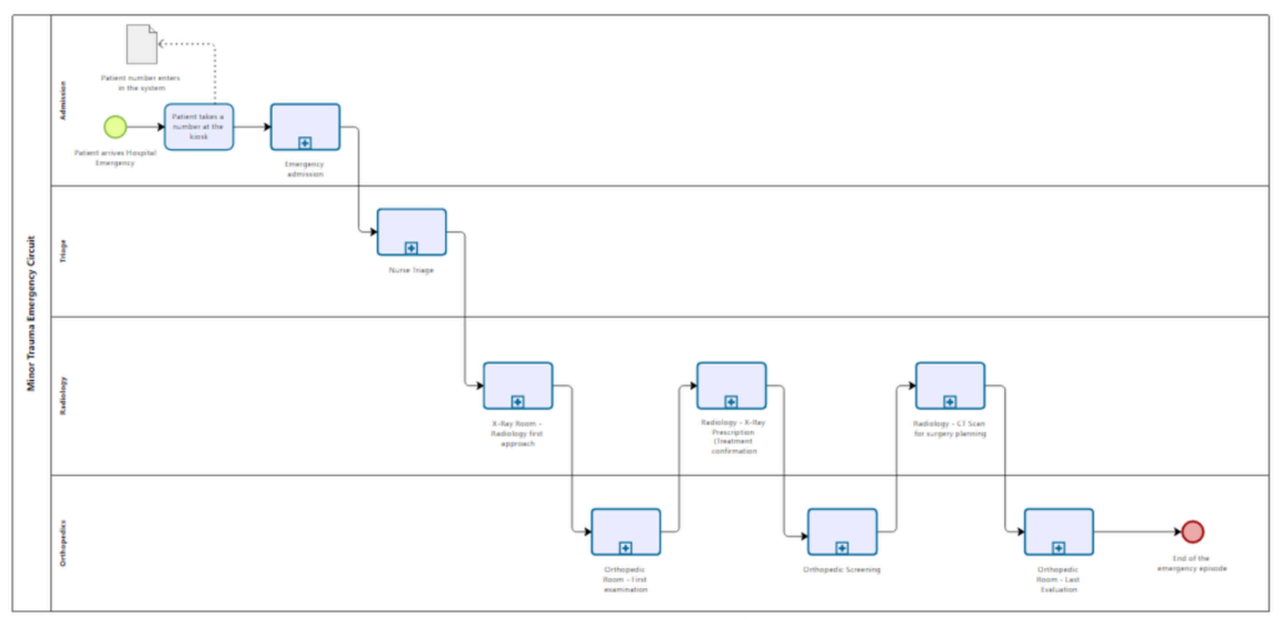
Quantitative Analysis: Discrete-Event Simulation (Time, Cost, Resources)

Redesign Strategy: Heuristics (Re-sequencing & Parallelism)



AS-IS Process Modeling



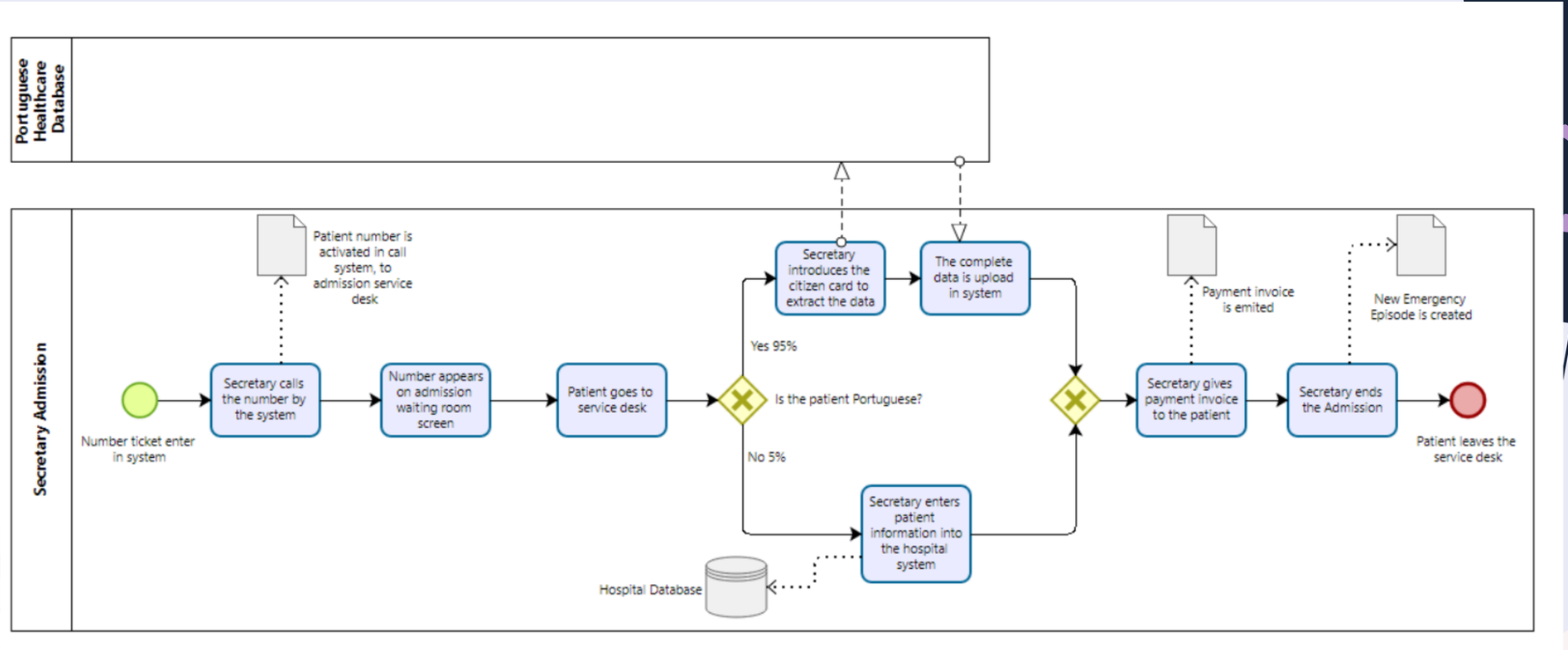
Overview: The process comprises 4 main lanes: Admission, Triage, Radiology, and Orthopedics

Level 1 Overview: The macro process flows through four main departments: Admission, Triage, Radiology, and Orthopedics.
Detailed Modelling: The process was decomposed into **8 specific sub-processes** to capture all operational details:

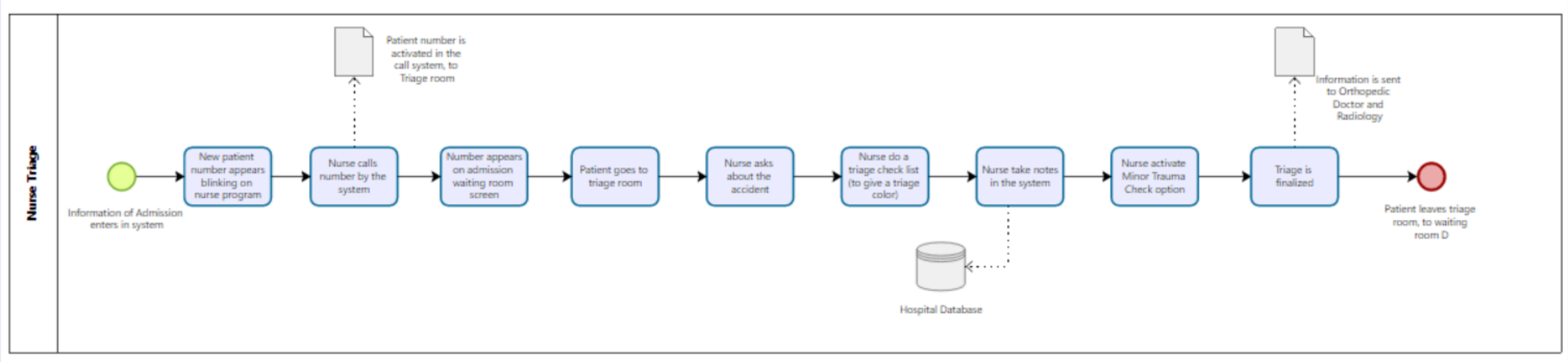
1. **Admission** (Registration & Administrative data).
2. **Triage** (Prioritization & MonoTrauma activation).
3. **Radiology:** First Approach (Initial X-Ray execution).
4. **Orthopedics:** First Evaluation (Diagnosis & Decision).
5. **Radiology:** X-Ray Prescriptions (Control exams for treatment confirmation).
6. **Orthopedic Screening** (Review of control images).
7. **Radiology:** CT Scan (For surgery planning).
8. **Orthopedics:** Final Evaluation (Discharge or Surgery Admission)

AS-IS Process Modeling

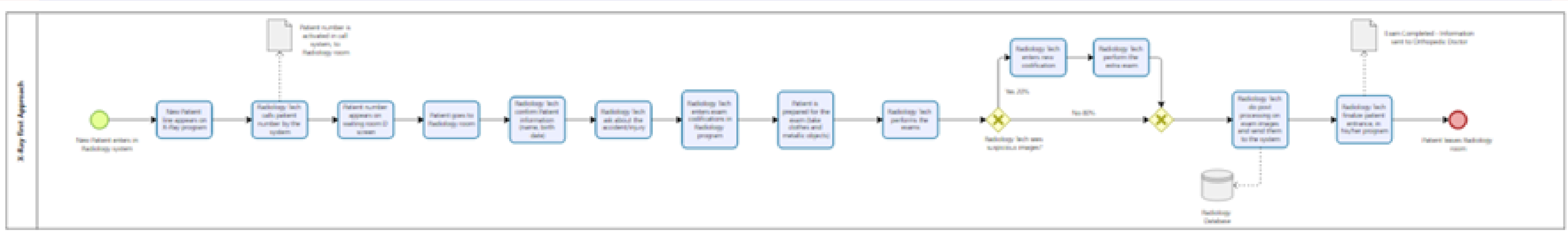
Admission sub-process



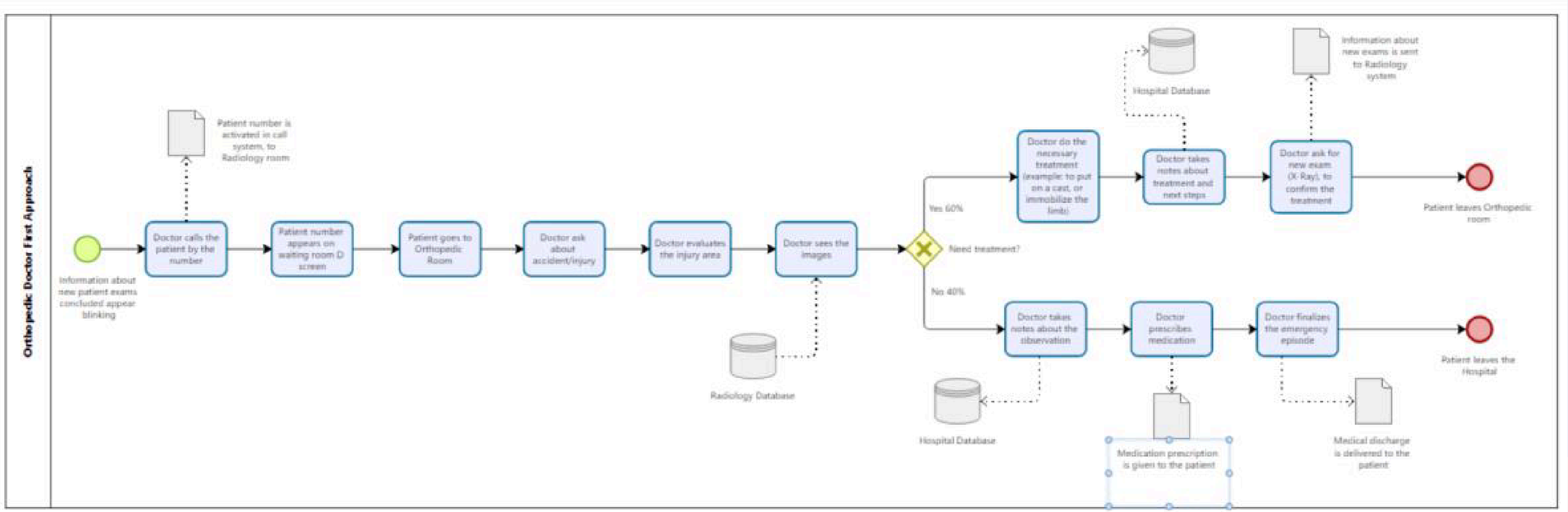
Triage sub-process



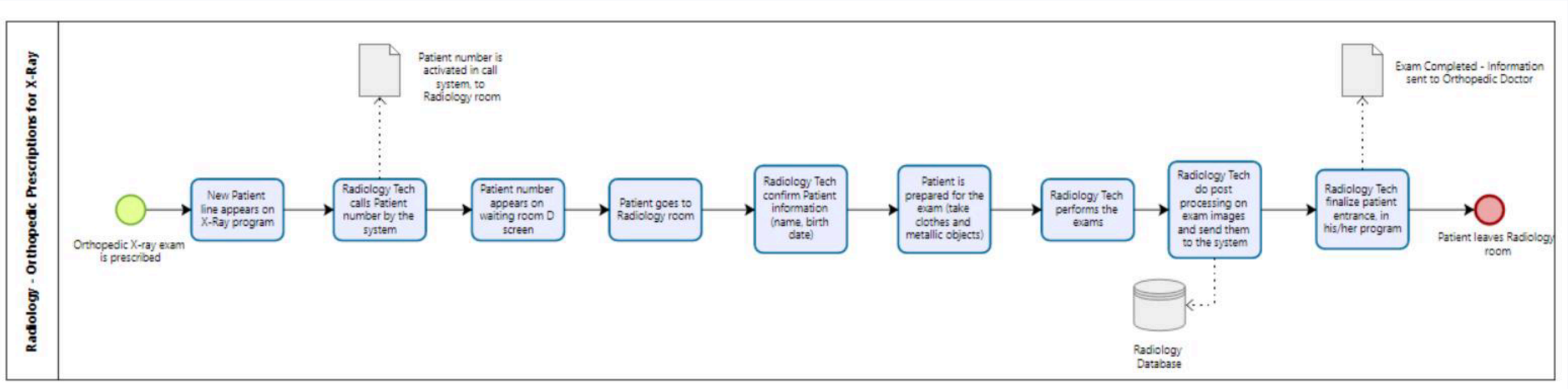
Radiology - first approach sub-process



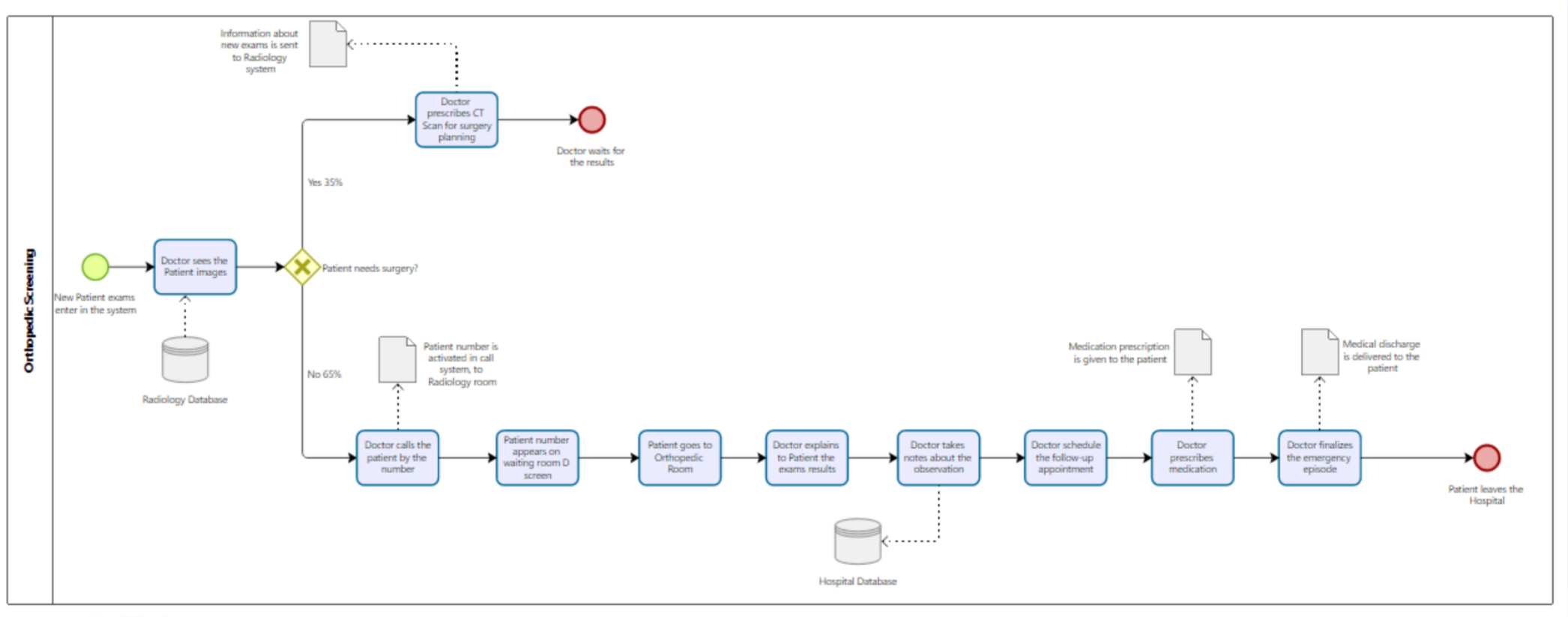
Orthopedics first evaluation sub-process



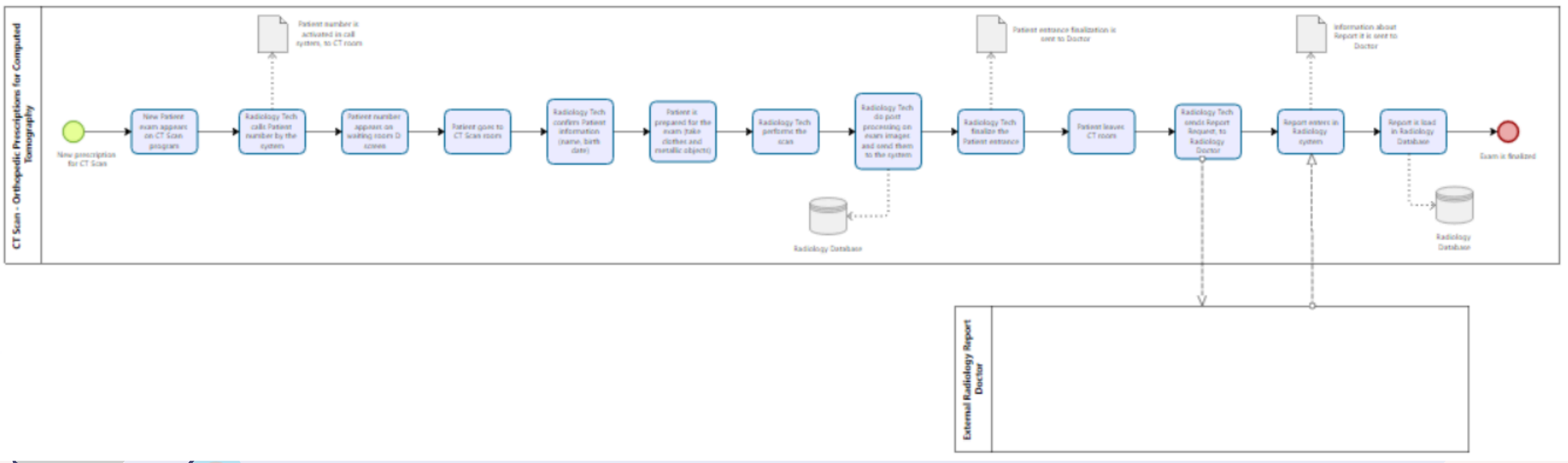
Radiology - X-Ray prescriptions sub-process sub-process



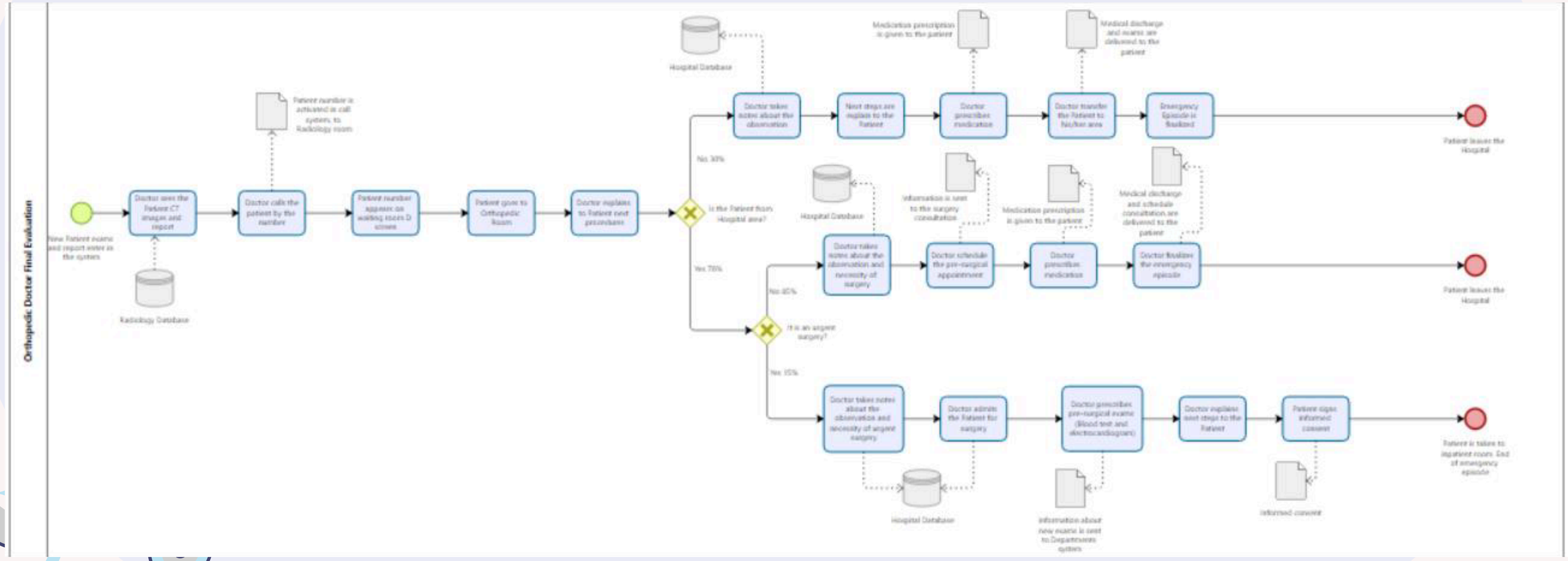
Orthopedic Screening sub-process



Radiology - CT scan (for surgery planning) sub-process



Orthopedy final evaluation sub-process



Qualitative Analysis (Issues Identified)

Value-Added Analysis: The process is patient-centered but contains Non-Value-Adding (NVA) operational waste

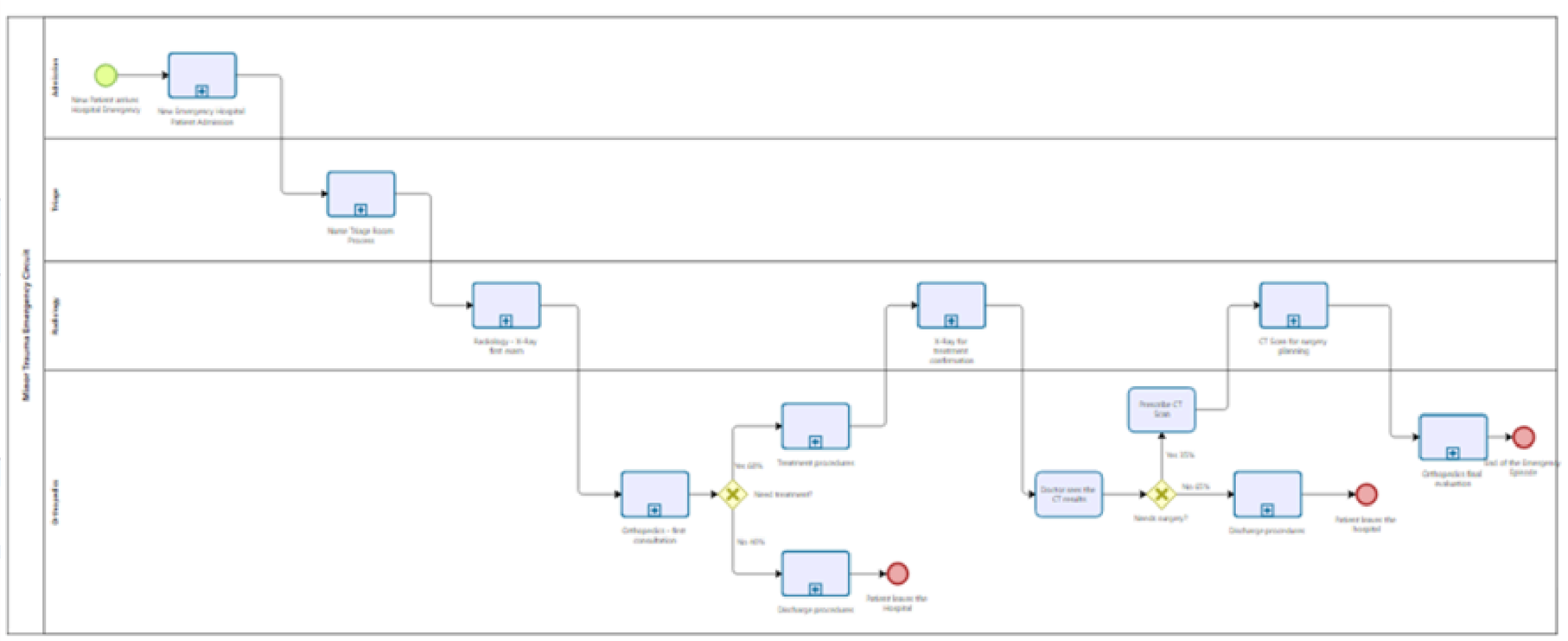
Key Issues (Issue Register):

- **Excessive Waiting Times**: Long waits between clinical stages.
- **Radiology Bottleneck**: Workload concentration and repetitive patient movements

- **External Dependency:** Delays due to external CT reporting
- **Patient Movement:** Patients cross departments 4-6 times per visit.

Quantitative Analysis (AS-IS Baseline)

Input Data:
Average of 70 patients/12h shift
Poisson Distribution
Mean arrival rate (λ approx 0.097)
;Mean inter-arrival time (1/λ) = 10.3



Quantitative Analysis (AS-IS Baseline)

Input Data:

4 What-if:

- - 1 Receptionist ✓
- - 2 Receptionists ✗
- + 1 Doctor ✓
- Elimination of taks ✓

Simulation Results:

- **Throughput:** Only 59 of 70 patients complete the process per shift.
- **Avg. Cycle Time**: 2h 14m 24s.

Resource Utilization: High strain on Orthopedists (86.77%) and Radiology Techs (73.26%); Low utilization for Receptionists (23.97%)

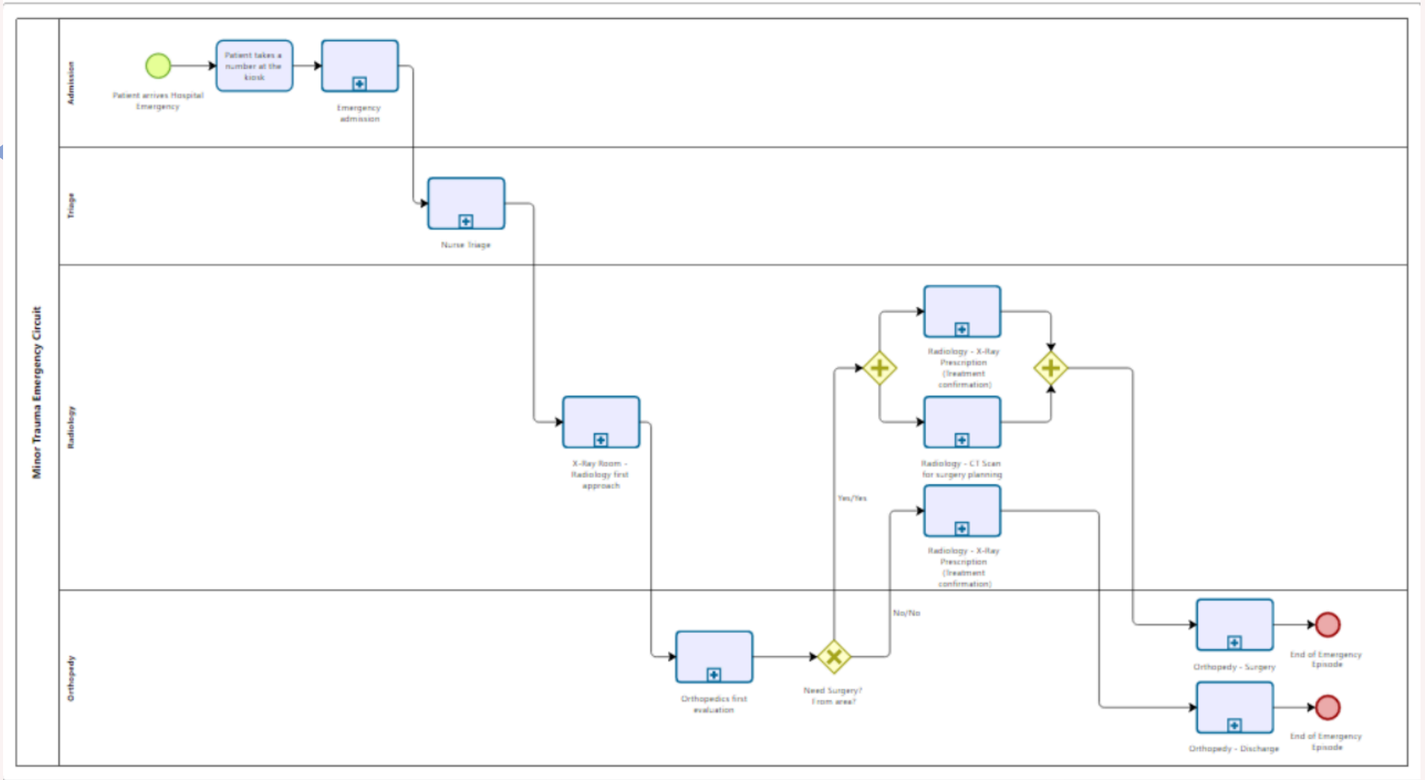
Redesign Heuristics (TO-BE)

Heuristics Applied:

Re-sequencing (Knock-out): Early "Hospital Area" check filters 30% of surgical cases, avoiding unnecessary planning steps .

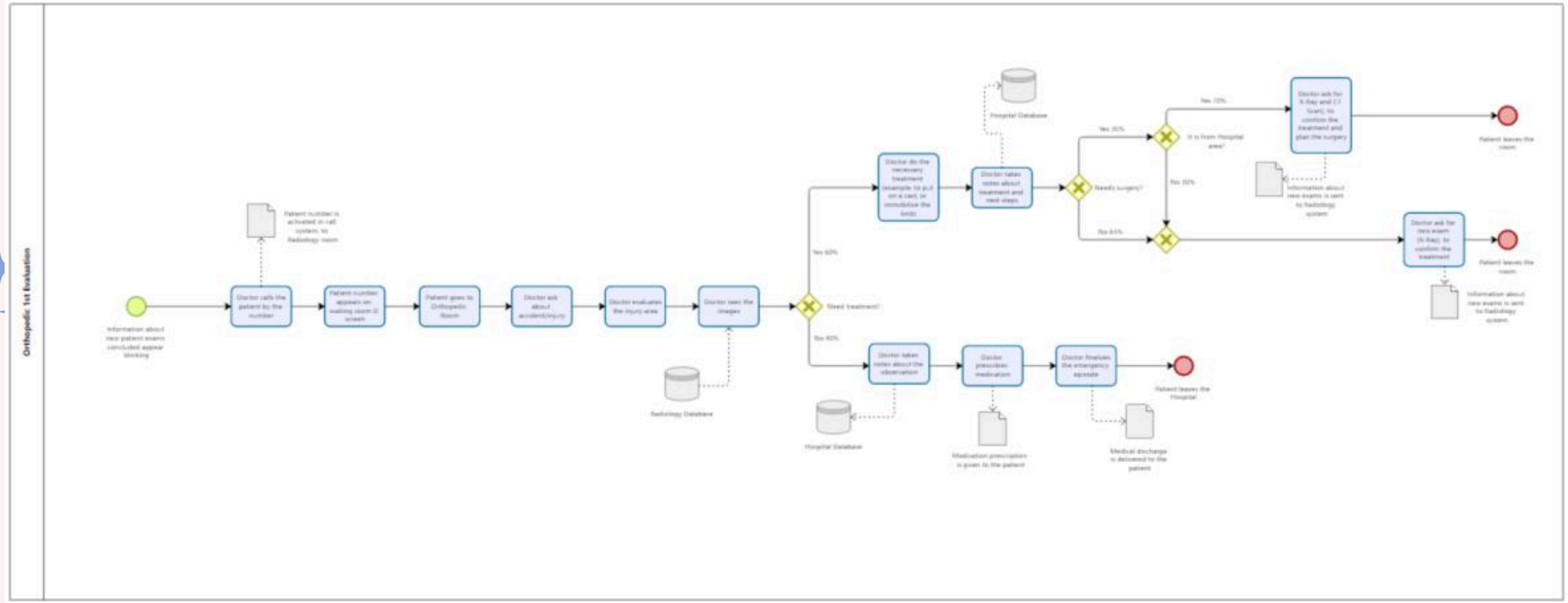
Re-sequencing (Decision): "Need Surgery?" check moved earlier to skip redundant X-Ray prescriptions .

Parallelism: X-Ray and CT scans now run in parallel for surgical patients, reducing lead time



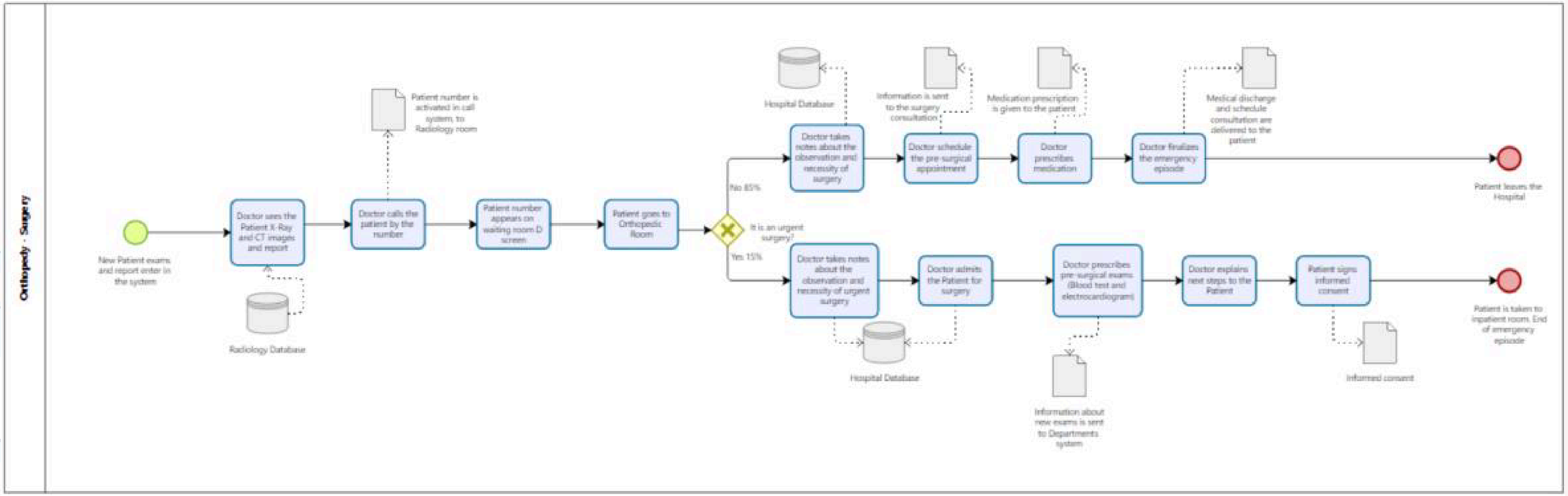
New Sub-processes for the TO-BE model

Due to these flow changes, 3 New Sub-processes were created for the TO-BE model



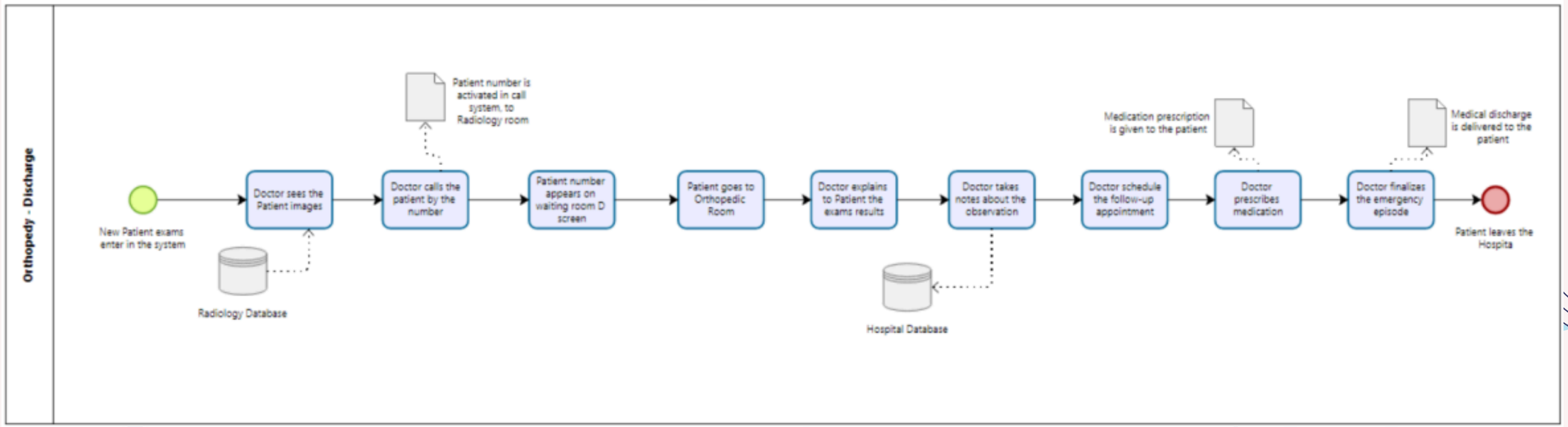
TO-BE Orthopedy First Evaluation

New Sub-processesfor the TO-BE model



TO-BE Orthopedy Surgery sub-process

New Sub-processesfor the TO-BE model



TO-BE Orthopedy Discharge sub-process

Final Comparison (AS-IS vs. TO-BE)

Cycle Time:

Reduced from 2h 14m (AS-IS) to 1h 14m (TO-BE). Improvement of > 1 hour

Throughput:

Increased from 59 to 68 patients completed per shif

Efficiency:

Optimized resource usage with better flow

Financial Evaluation

Total Cost per Shift:

- AS-IS: €956.20
- TO-BE: €968.25

Variation: Minimal increase of + €12.05 per shift.

Conclusion: The substantial gain in capacity and speed justifies the marginal cost increase.

Conclusion

BPM Success: The project successfully applied the BPM lifecycle to optimize the Minor Trauma Circuit.

Efficiency Gains: Achieved a ~50% reduction in cycle time and increased capacity through flow redesign and resource adjustment.

Value: Demonstrated that structured process analysis can significantly improve healthcare operations with minimal financial impact

Limitations

Academic Scope: Project limited to Analysis and Redesign phases; Implementation and Control were excluded.

Data Estimation: Simulation relies on estimated averages rather than empirical data.

Resource Focus: Model excludes shared resource constraints (interdependencies) with other hospital departments.

External Constraints: Contractual dependency on external CT reporting could not be altered